

Submitted by.....

1. In the circuit depicted in Fig. 1, i_x is determined to be 1.5 A, and the 9 V source supplies a current of 7.6 A (that is, a current of 7.6 A leaves the positive reference terminal of the 9 V source). Determine the value of resistor R_A .

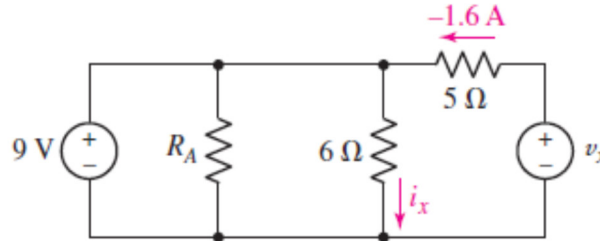


Fig. 1

2. Find the equivalent resistance R_{eq} in the network in Fig.2.

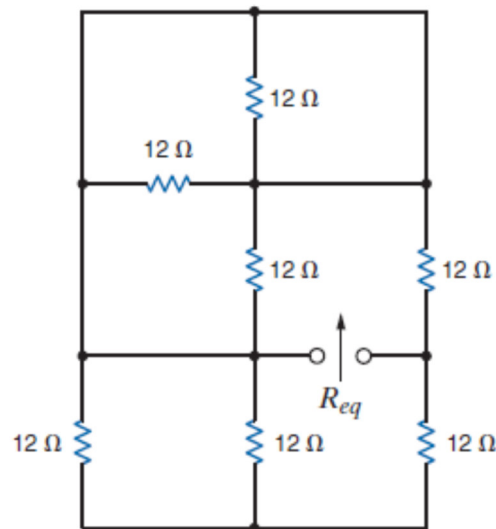


Fig. 2.

3. Determine the value of v_x as labeled in the circuit of Fig. 3.

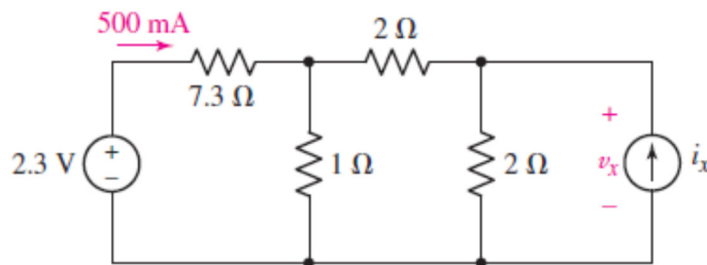


Fig. 3.

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4. Determine the power absorbed by the $15\ \Omega$ resistor in the circuit of Fig. 4.

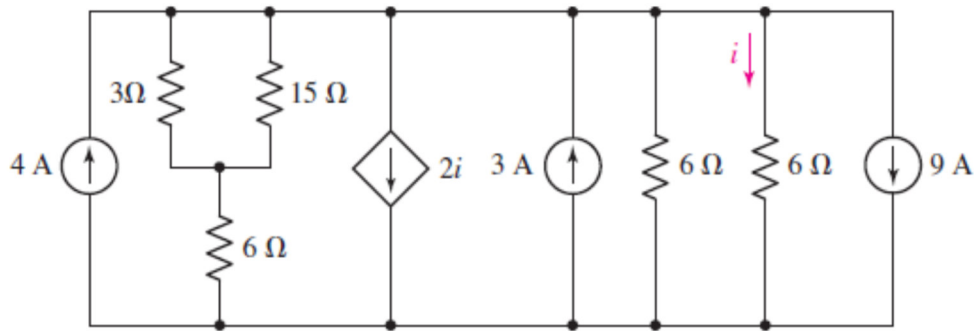


Fig. 4

5. Determine V in the circuit of Fig. 5.

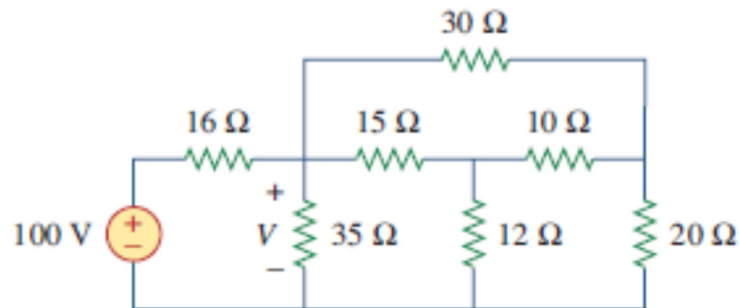


Fig. 5.

6. Determine the current labeled i in the circuit of Fig. 6 after first transforming the circuit such that it contains only resistors and voltage sources.

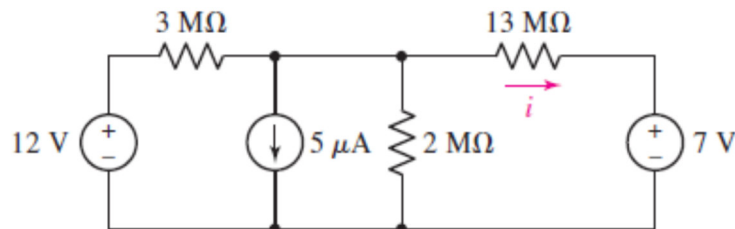


Fig. 6

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7. Employ superposition principles to obtain a value for the current I_x as labeled in Fig. 7.

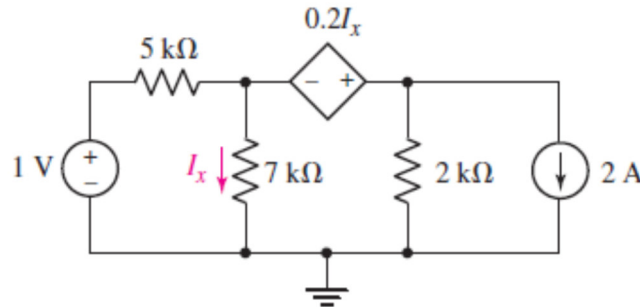


Fig. 7.

8. Determine the Thévenin equivalent of the network looking into terminal a-b. shown in Fig. 8 .

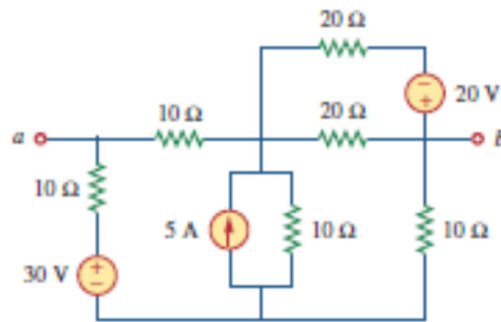


Fig. 8

9. Determine the Norton equivalent of the circuit drawn in Fig. 9 as seen by terminals a and b . (There should be no dependent sources in your answer.)

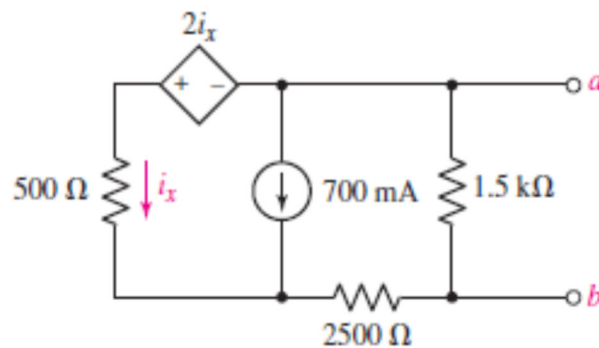


Fig. 9

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10. Find the value of R_L for maximum power transfer in the circuit of Fig. 10 and find the maximum power.

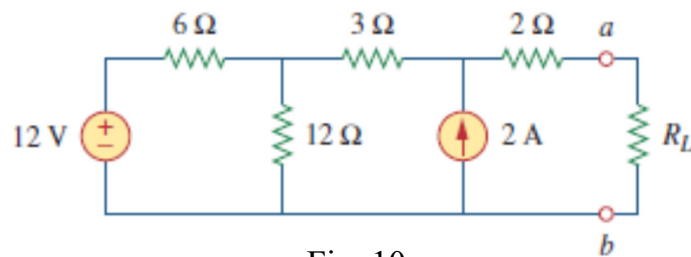


Fig. 10